Om Patel

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Education

Rutgers University–New Brunswick Sep 2025 – May 2027
Master of Science in Computer Science; GPA: 3.92/4.0 *New Brunswick, NJ*

Pandit Deendayal Energy University Jun 2021 – Jun 2025
Bachelor of Technology in Information & Communication Technology; CGPA: 8.93/10 *Gandhinagar, India*

Technical Skills

Languages: Python, C++, SQL, R, Java, TypeScript, JavaScript
Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, CNNs, GANs, U-Net, Computer Vision
LLMs & GenAI: LLM APIs (Llama 3.3, Groq), Prompt Engineering, Streaming Inference, Speech/TTS Pipelines
Data: Pandas, NumPy, SciPy, OpenCV, NLTK, Matplotlib, Spark, PostgreSQL, Redis, MongoDB, Qualtrics
MLOps & Deployment: AWS (EC2, S3, Lambda), Docker, GitHub Actions, FastAPI, Flask, Streamlit, REST APIs, Locust, Git

Experience

Research & Assessment Graduate Coordinator (Part-Time) Jul 2026 – Present
Rutgers University – Career Exploration and Success *New Brunswick, NJ*

- Develop and maintain the Python data pipeline behind an employer-facing dashboard showing career fair attendance by major to attract new recruiters (HTML/CSS, JSON); used by 10+ staff ahead of public launch
- Designed a rule-based ranking algorithm that scores student–mentor compatibility, producing ∼40 mentor–mentee pairs
- Map student engagement with Career Exploration and Success across majors in Python, surfacing low-engagement majors

Research Intern – Deep Learning Jan 2025 – May 2025
Pandit Deendayal Energy University *Gandhinagar, India*

- Engineered a Dual-Stream U-Net for satellite change detection, reaching 89% IoU and a 17% higher deforestation F1
- Beat FCN-8 baselines on land-cover reconstruction with optimized U-Nets: +22% PSNR, 0.85 SSIM
- Automated data pipelines for 10,000+ geospatial images (TensorFlow, PyTorch), cutting preprocessing latency by 40% and improving generalization through data augmentation

Embedded Systems Intern May 2024 – Jul 2024
Candore Bioscience *Ahmedabad, India*

- Developed a sensor-based inference algorithm that estimates cryogenic liquid levels from noisy thermal and ultrasonic signals (ESP32 MicroPython, Arduino Embedded C), improving detection reliability by 40%
- Applied Steinhart-Hart regression and exponential smoothing to temperature signals, enabling stable readings to −23°C
- Programmed real-time data acquisition for sensor calibration, signal filtering, and anomaly-resistant measurement

Projects

Composur | *Llama 3.3 70B (Groq), Next.js, TypeScript, PostgreSQL, AWS S3* composur.tech

- Launched a live mock interview platform where Llama 3.3 70B (via Groq) conducts voice-based technical interviews, generating adaptive follow-up questions and structured performance evaluations
- Created a real-time voice pipeline chaining speech recognition, streaming LLM responses, and low-latency text-to-speech (ElevenLabs), with role-specific prompt sets for 28 interview domains
- Deployed a TypeScript/PostgreSQL backend on Vercel with S3 uploads, rate limiting, and GitHub Actions CI testing

SRGAN Image Super-Resolution | *PyTorch, GANs, CNNs, VGG19, OpenCV, LPIPS*

- Implemented an SRGAN (generator–discriminator) in PyTorch for 4× upscaling, with paired-data loading and checkpointing
- Trained in two stages: MSE pre-training, then adversarial fine-tuning with VGG19 perceptual and total-variation losses
- Integrated PSNR and LPIPS evaluation to measure pixel-level fidelity and perceptual similarity to ground truth

EventPass | *Python, FastAPI, PostgreSQL, Redis, Docker, Locust* github.com/Ompatel179/Eventpass

- Built and deployed a QR-code event check-in API for student clubs (FastAPI, PostgreSQL, SQLAlchemy, Redis, JWT auth) with Docker and GitHub Actions CI on Render and Neon
- Guaranteed exactly-once check-in with a database unique constraint: 0 duplicates in 2,862 concurrent scans
- Load-tested with Locust and fixed a deadlock that froze the API at 13 req/s, reaching ∼950 req/s with 0 errors on one worker; an index cut a 1M-row query from 16.2 ms to 0.08 ms

Publications

Harnessing Generative Adversarial Models to Enhance Satellite Imagery Quality Aug 2025

- Published at the 2025 IEEE International Conference on Emerging Technologies and Applications (MPSec ICETA): SRGAN/ESRGAN framework for satellite image reconstruction achieving 34.55 dB PSNR and 0.08 LPIPS